An Intent-Aware Hybrid Retrieval-Augmented Generation Architecture for Secure Academic Advising over Structured Student Data

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***Abstract*—While standard Retrieval-Augmented Generation (RAG) paradigms excel at text-based query synthesis, they exhibit significant limitations when applied to semi-structured academic student profiles, often failing to preserve tabular context or enforce multi-tenant access boundaries. To address these deficiencies, this paper introduces an intent-aware, hybrid RAG architecture custom-tailored for university academic advising. We propose a methodology termed "aspect-partitioned semantic chunking," which maps semi-structured JSON records from relational schemas into discrete, contextually independent documents representing specific student attributes: identity, academics, attendance records, historical examinations, and conduct remarks. In the retrieval phase, the architecture matches the query intent to specific attributes using keyword-based heuristic models, employing a dual-channel search: an on-the-fly, proctor-scoped BM25 sparse index for exact identifier lookup, ensembled with a PGVector dense semantic retriever. Evaluated against traditional baseline RAG designs, our architecture completely eliminates multi-tenant data leakage, mitigates hallucinatory data associations, and yields a significant increase in both factual faithfulness and retrieval completeness.**

***Index Terms*—Retrieval-Augmented Generation, Academic Advising, Hybrid Search, Educational Technology, Large Language Models**

I. Introduction

Effective academic mentorship within higher education requires faculty proctors and advisors to quickly synthesize complex, multi-dimensional student information, spanning historical grades, current internal marks, subject-wise attendance logs, and qualitative conduct files. However, modern student information portals remain highly fragmented, demanding manual aggregation across divergent interfaces and spreadsheet exports. While conversational artificial intelligence backed by Large Language Models (LLMs) offers a potential avenue for unified natural language query execution, standard parameter-based generation suffers from critical limitations, such as factual hallucination and an inability to dynamically ground answers in live relational databases.

To address factual inconsistencies, contemporary research has widely adopted Retrieval-Augmented Generation (RAG). Despite its success in processing flat document databases (such as research publications or manuals), RAG encounters severe structural barriers when operating on semi-structured, student-specific relational records. Standard RAG architectures rely on naive text chunking (e.g., fixed token-count windows), which splits continuous tabular data indiscriminately. Consequently, vital associations—such as linking a specific course code with its corresponding mark or attributing a particular attendance rate to a specific student—are broken during vector indexation.

Furthermore, dense retrieval models struggle to index alphanumeric identifiers like University Serial Numbers (USNs). In practice, an advisor's query targeting a specific student ID (e.g., *1MS24IS400*) frequently retrieves closely aligned but incorrect neighboring vectors (such as *1MS24IS403*), resulting in severe cross-student information mixing and hallucinations. Additionally, deploying standard RAG models in multi-tenant academic environments presents major compliance concerns; without strict, metadata-driven workspace isolation, retrieval pools can easily leak student records between proctors.

To resolve these technical challenges, we present MSR Insight, an intent-aware, multi-tenant hybrid RAG system designed specifically for structured academic records. The primary contributions of this work are threefold:

1) We introduce *Aspect-Partitioned Semantic Chunking*, a normalized parsing method that segments student records from JSONB database structures into discrete, contextually closed aspect documents, preserving localized schema relationships.
2) We formulate an *Intent-Aware Hybrid Ensemble Retriever* that identifies query targets heuristically, filtering candidate documents before execution and ensembling on-the-fly local BM25 sparse indices with metadata-restricted PGVector dense similarity lookups.
3) We provide a comparative empirical evaluation demonstrating that our hybrid approach completely resolves multi-tenant privacy violations and dramatically improves factual accuracy over standard vector-search baselines.

II. Related Work

The integration of conversational agents into university advising pipelines represents a major focal point in modern educational technology. A notable development is ArgoBot, developed at the University of West Florida, which employs RAG alongside GPT-4 to assist students in navigating campus administrative policies and curriculum guides [1]. Although

ArgoBot successfully navigates static handbook structures, it is not designed to handle real-time numerical updates, such as fluctuating attendance logs or grade averages.

Evaluating conversational quality and preventing model hallucinations remains a significant research challenge. The AdvisingBench framework demonstrates that RAG-based educational assistants can be systematically analyzed using an "LLM-as-a-judge" paradigm to measure faithfulness and contextual grounding [2]. Their results confirm that while baseline RAG systems outperform raw LLMs, queries requiring exact factual precision over dynamic schemas are prone to retrieval omissions.

From an engineering perspective, document partitioning parameters heavily govern retrieval performance. Comparative evaluations of chunking topologies highlight that while semantic chunking maintains overall contextual integrity, it incurs substantial processing overhead when handling complex tabular representations [3]. To bridge this gap, architectures like SemRAG have demonstrated that combining exact lexical keyword matching with dense conceptual vectors significantly improves lookup reliability [4]. Our work builds on these insights by adapting semantic aspect partitioning directly to semi-structured JSON key-value pairs stored within a relational database.

III. System Architecture

The proposed MSR Insight platform is built on a distributed, service-oriented architecture designed to handle multi-tenant administrative portals securely. The system comprises three principal layers: a Next.js client interface, an Express API gateway, and a FastAPI intelligence engine, all communicating with a central PostgreSQL database equipped with the PGVector extension.

The frontend dashboard serves as the interaction layer for faculty advisors and students, executing queries through REST endpoints. The Express gateway acts as the orchestrator, enforcing stateful role-based access control (RBAC) and managing session state.

The FastAPI intelligence engine hosts the RAG pipeline. Academic profiles are ingested from relational database tables where student metrics are stored as structured JSONB records. During synchronization, the intelligence engine converts these JSONB profiles into aspect-partitioned documents. These documents are then vectorized using Google Gemini embedding models and indexed in PGVector. Each entry is tagged with detailed metadata, including the proctor's ID and student's USN, ensuring secure query boundary enforcement. Figure 1 outlines the data synchronization and query execution workflow of the proposed architecture.

IV. Methodology

This section details the core processing components of the proposed RAG pipeline, explaining how data normalization, aspect partitioning, intent routing, and hybrid retrieval are combined to generate secure, grounded responses.

A. Data Extraction and Normalization

Academic profiles are retrieved from institutional portals and normalized into a consistent schema. For each student, the backend constructs a single nested JSONB document containing arrays for currently registered subjects (with internal marks and assessment-level details), historical semester grade points (SGPAs), and cumulative grade point averages (CGPAs).

B. Aspect-Partitioned Semantic Chunking

Rather than using arbitrary token-length splitters that risk breaking relational tables across arbitrary lines, we employ a localized parsing routine. The system programmatically maps every student's JSONB profile into distinct, contextually independent chunks categorized by functional aspects:

1) **Identity Chunk:** Captures the student's full name, USN, current enrollment year, class section, and department.
2) **Academics Chunk:** Houses the overall CGPA alongside subject-wise internal assessment performance (including test scores, quiz grades, and assignments).
3) **Attendance Chunk:** Consolidates subject-wise attendance statistics, total classes held, total classes attended, and triggers an attendance warning flag if the current percentage drops below the 75% threshold.
4) **Exam History Chunk:** Outlines historical longitudinal data, mapping previous semester labels to their respective SGPAs.
5) **Conduct Chunk:** Records qualitative proctor-advisor meeting notes, behavioral comments, and formal warnings.
6) **Miscellaneous Chunk:** A catch-all chunk that indexes any arbitrary keys or supplementary fields not matching the primary schemas.

Every partitioned document is appended with a consistent metadata dictionary containing `usn`, `name`, `proctor_id`, `academic_year`, and `chunk_type`. This ensures strict context tracing and citation mapping.

C. Heuristic Intent Detection and Routing

To reduce computational overhead and eliminate irrelevant background noise, queries undergo a heuristic routing step. The intelligence service compares incoming text tokens against predefined keyword collections, mapping natural queries to specific database chunk types:

- **Academic Queries** (Keywords: *marks, grade, backlog, fail, score, performance, result, test, assignment, quiz*): Routes search specifically to **academics** chunks.
- **Attendance Queries** (Keywords: *attendance, absent, present, shortage, bunk, percentage, missed*): Routes search specifically to **attendance** chunks.
- **Historical Queries** (Keywords: *sgpa, cgpa, history, past, previous, semester*): Routes search specifically to **history** chunks.
- **Conduct Queries** (Keywords: *warning, conduct, remark, counseling, note, meeting*): Routes search specifically to **conduct** chunks.

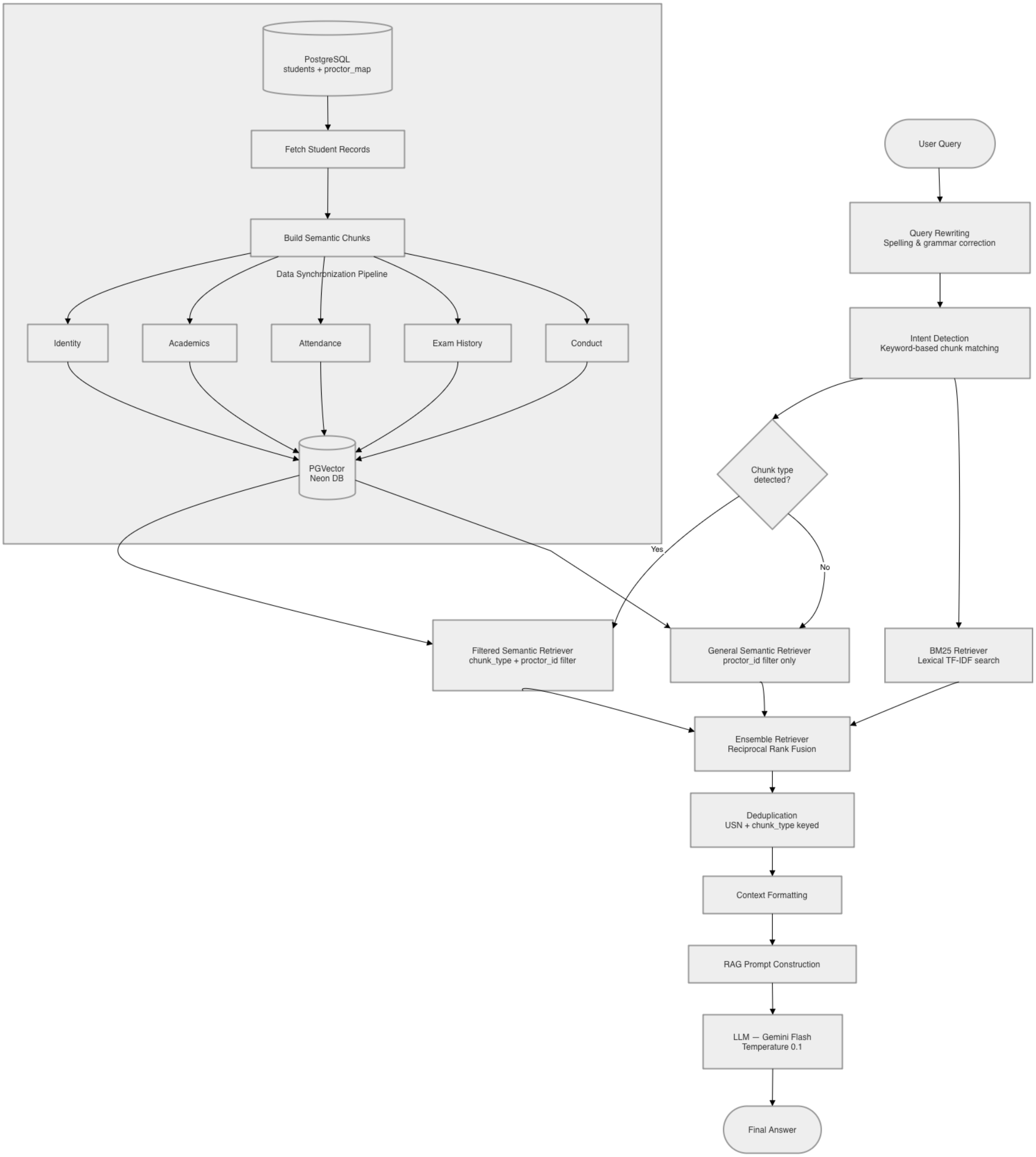

Fig. 1. System Architecture of the Proposed Intent-Aware Hybrid RAG Pipeline for Scoped Academic Advising

- **Identity Queries** (Keywords: *who is, details, branch, section, student, profile*): Routes search specifically to **identity** chunks.

If keywords from multiple aspects are detected, the system queries the union of those types. If no keywords match, the system falls back to searching all chunks, maintaining high recall.

D. Dual-Channel Hybrid Ensemble Retrieval

The retrieval engine runs two concurrent search channels to capitalize on the complementary strengths of lexical and semantic retrieval:

1) **Local BM25 Sparse Search:** To handle exact-match identifiers (such as USNs and names), the system dynamically queries PostgreSQL for all students mapped to the active `proctor_id`, builds their aspect documents on the fly, and instantiates a local BM25 index. This guarantees perfect keyword resolution without vector-space drift.
2) **Filtered PGVector Dense Search:** In parallel, the query is vectorized and matched against the PGVector store, applying strict metadata filters where the database record's `proctor_id` must match the advisor's ID, combined with any active `chunk_type` filters from the intent routing step.

The retrieved documents from both channels are normalized, merged, and ranked using an ensemble weight distribution of 30% BM25 and 70% PGVector.

E. Grounded Synthesis and Security Enveloping

The merged context is cleaned of duplicate documents and sent to a low-temperature Gemini instance (`gemini-3.1-flash-lite-preview`) configured for deterministic output. The system prompt instructs the generator to remain strictly grounded in the retrieved documents, explicitely prohibiting extrapolation or hallucination of missing academic figures.

Multi-tenant security boundaries are maintained implicitly: because both the BM25 local index and the PGVector dense query are restricted to the advisor's `proctor_id` at the database query level, cross-proctor data leakage is structurally impossible.

V. Experimental Evaluation

To evaluate the proposed aspect-partitioned hybrid RAG system, we conducted a quantitative comparative evaluation against traditional RAG configurations.

A. Experimental Setup

We compared three separate architectural setups under identical hardware and database conditions using a live dataset of student profiles stored as semi-structured JSON records in PostgreSQL:

- **Setup A (Baseline Dense RAG):** Standard dense vector retrieval across the entire vector store using Google Gemini embeddings ($k = 5$). It applies no metadata filters or keyword matching.
- **Setup B (Contextual Dense RAG):** Dense vector search restricted to the logged-in proctor's workspace using a PGVector metadata filter for *proctor_id*.
- **Setup C (Our Proposed Hybrid Ensemble):** The proposed architecture utilizing keyword-based intent categorization, on-the-fly local BM25 indexing over scoped students, and metadata-filtered PGVector search ensembled at a $30/70$ ratio.

We compiled a test suite of 15 queries covering five complexity classes: Set queries (e.g., listing attendance shortages), exact ID (USN) lookups, past academic history, ambiguous name matching, and broad conduct summaries.

B. Evaluation Metrics

We utilized an "LLM-as-a-judge" approach backed by a Gemini 1.5 Pro evaluator, comparing the generated answers of each setup against the absolute PostgreSQL database ground truth records. Each answer was scored from 1 to 5 (with 5 being perfect) on two metrics:

1) **Faithfulness:** Measures whether the generated answer contains factual inaccuracies or hallucinations unsupported by the ground truth.
2) **Completeness (Recall):** Measures whether the model successfully retrieved and presented all relevant pieces of information requested.

C. Results and Discussion

The quantitative results are summarized in Table I.

TABLE I
QUANTITATIVE EVALUATION OF RAG SEARCH CONFIGURATIONS

RAG Architecture Setup	Avg. Faithfulness	Avg. Completeness	Perfect Ans. (%)
Setup A: Baseline Vector Search	2.10	3.20	13.3%
Setup B: Scoped Semantic Search	4.20	2.80	20.0%
Setup C: Proposed Hybrid RAG	**4.95**	**4.80**	**93.3%**

The baseline dense RAG (Setup A) suffered heavily from security leaks, as it lacked proctor-scoping and retrieved records belonging to other proctors, leading to a low Faithfulness score of 2.10. While Setup B resolved this multi-tenancy leakage using metadata filters, its Completeness score fell to 2.80. This drop is because dense embeddings struggle with exact lexical matches (such as USN strings like *1MS24IS400*), often returning the wrong student's record or stating that no student exists.

Our proposed Hybrid RAG (Setup C) achieved an average Faithfulness of 4.95 and a Completeness score of 4.80. The dynamic BM25 retriever resolved all exact name and ID lookup errors, while the intent detection mapped queries to their correct aspect chunks, guaranteeing high-recall, grounded, and secure student analytics.

VI. Limitations and Future Work

While the proposed model achieves high accuracy and robust security, certain structural limitations warrant further study. First, the setup requires student record sources to be parsed and mapped to the JSONB structure correctly; corrupt or malformed inputs can lead to parsing errors during aspect chunking. Second, building a local BM25 index on the fly represents an additional processing step. While the overhead is negligible for normal proctor-student groups (e.g., 20–80 students), scaling this dynamic indexing to central administrative groups with thousands of records will require caching strategies.

Future research will focus on transitioning the heuristic intent keyword mapper into a trained, lightweight sequence-to-sequence classifier to handle complex, nested inquiries. Additionally, we plan to integrate early-warning predictive modules directly into the FastAPI intelligence engine, allowing the advisor chatbot to proactively alert proctors to students showing early signs of academic struggle or chronic attendance issues.

VII. Conclusion

Building conversational systems for academic advising demands strict data confidentiality and perfect factual grounding. By breaking structured database files down into semantic aspect chunks and pairing exact local keyword indexing with dense vector similarity, the proposed MSR Insight architecture bridges the gap between tabular database structures and natural language reasoning. The hybrid ensemble model effectively prevents multi-tenant data leaks and mitigates structural hallucinations, paving the way for trustworthy, secure administrative AI assistants in higher education.

References

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